ChatGPT

Executive Summary

Polymetallic mineral deposits – especially deep-sea **manganese nodules**, **seafloor massive sulphides (SMS)**, and onshore **nickel–cobalt laterites** and **Cu–Co sulphides/veins** – hold very large stores of critical metals (Ni, Co, Cu, Mn, Zn, Pb, Ag, Au and REEs). The **Clarion-Clipperton Zone (CCZ)** in the Pacific (ISA, international waters) hosts the world's largest nodule resource (~21 billion tonnes【54†L42-L47】), rich in Ni, Cu, Co and Mn. Other major nodule fields lie in the **Central Pacific (Peru Basin)** and **Central Indian Ocean**【81†L198-L205】. Seafloor SMS occur along mid-ocean ridges and volcanic arcs; e.g. **West Pacific back-arc basins (Papua New Guinea, Solomon Sea)** and the **Mid-Atlantic Ridge** are known sites, with tens of thousands of km of ridge hosting ~1,000 hydrothermal vents (~100 with substantial sulphide mineralisation【88†L64-L72】【86†L75-L82】). Land-based polymetallic deposits include tropical **Ni–Co laterites** (Indonesia, Philippines, Australia, Cuba, New Caledonia, etc.), and major Cu–Co sulphide belts (Katanga, DRC; Norilsk, Russia; Sudbury, Canada). Using current metal prices (mid-2026: Ni ~$18,000/t【40†L244-L249】, Cu ~$10,000/t, Co ~$42,000/t【43†L132-L139】, Zn ~$3,500/t【90†L52-L60】, Ag ~$22,000/kg, Au ~$64,000/kg), CCZ nodules (~1.3% Ni, 1.0% Cu, 0.25% Co by mass) yield roughly **$0.9K per tonne** of raw nodules, whereas dense sulphide ores or crusts can exceed **$500–800K/t** due to high metal grades. Accessibility (remote abyssal depths or onshore jungles), regulatory regime (ISA vs national EEZ), environmental sensitivity and extraction cost vary widely.

We have identified the top 15 high-potential regions worldwide (ranked below) by combining geology, metal endowment and economic factors. A summary comparison is shown in the tables below. Key drivers include *deposit size and grade* (e.g. CCZ's 21 Gt of polymetallic nodules vs. tens of Mt in even largest land deposits), *jurisdiction* (international seabed vs. national EEZ), and *extraction cost/technology*. High-confidence prospects (e.g. Katanga, Norilsk, Sudbury, Indonesia) are well-studied and already producing, whereas lower-confidence targets (deep-sea sites, underexplored terrain) are richer but riskier. **Data gaps** (such as incomplete tonnage estimates for many SMS sites, or environmental impacts on deep ecosystems) and price volatility are noted where relevant. All metal prices are as of late 2026.

【32†embed_image】 *Figure: Major polymetallic nodule exploration claims in the Clarion-Clipperton Zone (Pacific Ocean)*. The CCZ spans ~6 million km^2 at 4–6 km depth【8†L25-L29】【54†L42-L47】. Key contract areas (colours) are held by various states/companies (refer ISA).

Deposit Types and Geology

- **Polymetallic nodules (manganese nodules)**: Spherical to discoidal concretions (2–10 cm) of Mn–Fe hydroxides (≥25% Mn by weight) that accrete ultra-slowly on abyssal plains【52†L739-L748】【81†L198-L205】. Formed by precipitation of metals (hydrogenetic/diagenetic processes) on any nucleus over millions of years【52†L749-L757】【52†L758-L765】. Key tectonic settings are old, slowly accumulating, sediment-starved abyssal plains at 4–6 km water depth (optimal at <5 kg nodules/m^2【81†L219-L224】). Major fields: Pacific CCZ, Peru Basin, Central Indian Basin【81†L198-L205】. Typical grades (CCZ average) are ~1.1–1.5% Ni, 0.8–1.2% Cu, 0.2–0.25% Co by mass, plus ~23% Mn【52†L752-L760】. They also contain significant Li, Y and REEs【52†L758-L765】. For example, Clarion-Clipperton has ~21×10^9 t nodules【54†L42-L47】. At mid-2026 metal prices, one tonne of typical nodules (Ni, Cu, Co, Mn only) is worth roughly USD$0.9K. Accessibility is challenging: depths ~4000–6000 m and often >1000 km from shore imply very high extraction costs. Legally these lie in "the Area" under ISA jurisdiction (17 contractors hold

exploration licenses there【8†L35-L39】【98†L232-L240】). Environmental sensitivity is high – >90% of abyssal species may be unknown【8†L25-L29】, and seabed disturbance (nodule removal) has long-lasting impacts. Technology (deep-sea mining vessels, collector vehicles) exists but has not yet been proven at scale. Exploration is advanced (numerous contract areas assessed) but no commercial production yet.

- **Seafloor massive sulphides (SMS)**: Hydrothermal vent deposits (analogous to onshore volcanic massive sulphides) precipitated from hot fluids on mid-ocean ridges, back-arc basins and volcanic arcs【88†L64-L72】. They typically form chimney-like structures or mounds (meters to hundreds of meters across) on the seafloor (depths ~1–3 km). SMS are rich in base metals: often tens of % Cu and Zn, with 1–10 g/t Au and ~100 g/t Ag【88†L64-L72】. For example, Nautilus Minerals' Solwara-1 (Bismarck Sea) was ~5–6% Cu, ~8% Zn, 19 g/t Au and 255 g/t Ag. The global ridge system (~60,000 km) and arcs host ~1000 active hydrothermal sites (only ~100 with commercial-grade sulphides)【88†L72-L81】. About 40% of known SMS deposits are in national EEZs (within 200 nm)【88†L72-L81】 – e.g. Papua New Guinea, Tonga, Japan, the Atlantic margin, etc. Size: individual SMS deposits are typically tens of kt to a few Mt of ore (much smaller than land analogues). At ~$10K/t Cu and ~$3.5K/t Zn, a 1 Mt deposit at 5% Cu+5% Zn is worth on the order of $900M. Jurisdiction and technology: several SM Sprojects have reached pilot/exploration (e.g. Solwara-1 had a mining license under PNG's EEZ rules but was halted by commercial/technical issues). Deep-sea remotely-operated mining equipment is under development. Environmental risk is major (unique vent ecosystems). Known discoveries: TAG, Logatchev (Atlantic), Okinawa Trough, Loki's Castle, Lucky Strike, etc., but no production to date【88†L72-L81】【86†L75-L82】.

- **Cobalt-rich ferromanganese crusts**: Dense Mn–Fe oxide crusts (1–25 cm thick) on seamount flanks (400–2500 m depth). They contain Co (0.4–1%) and Ni (~0.5%) plus Te, Mo, REEs【52†L769-L778】【52†L783-L790】. Global targets are Pacific and Indian seamounts. Crusts grow slowly (1–5 mm/Myr), so only thin, but high Co content can make them attractive. They are less studied than nodules but ranked high for Co. Exploitation would involve dredging seamount slopes (very niche). We mention them as potential Ni–Co sources but they are not yet prioritized regionally in top-15 (more speculative).

- **Polymetallic laterites (land)**: Tropical weathering profiles on ultramafic rocks producing **nickel–cobalt laterite** ores. These occur in belts in Indonesia (Sulawesi, Halmahera), Philippines, New Caledonia, Australia (Queensland), Cuba, Brazil, parts of Africa (Gabon, Tanzania) and elsewhere. These laterites typically grade 0.8–1.5% Ni and ~0.03–0.2% Co. Their sheer volume is large: Indonesia alone (~22% of global Ni reserves) produced ~50% of world Ni (2022)【76†L255-L258】. For instance, Indonesia's reserves are tens of million tonnes Ni (40–60% of world laterite reserves【76†L255-L258】). At ~$18K Ni and ~$42K Co, 1 tonne of 1.3% Ni + 0.05% Co ore is ~$0.25K. However, mining laterites is comparatively mature (open-pit operations, HPAL processing), though environmentally destructive (deforestation, tailings). Legally straightforward (onshore mining law), but social and ESG permitting can be challenging. These are proven resources (Indonesia, Philippines, New Caledonia have large operating mines) so confidence in resource and logistics is high.

- **Polymetallic veins/massive sulphides (land)**: Onshore analogues of polymetallic sulphide and vein deposits (e.g. volcanogenic massive sulphides, orogenic Au–Ag–Cu veins, SEDEX, etc.). Many large base-metal provinces (e.g. Katanga, Polish/Czech Ore Mountains, Cypriot VMS, etc.) include Ni, Cu, Co, Zn, Pb, Ag, Au. We highlight Katanga (Cu–Co sedimentary-exhalative deposits) and Norilsk/Sudbury (magmatic Ni–Cu–PGE) as top regions. Grades vary (Katanga ores ~2–5% Cu, 0.2–1% Co; Norilsk sulphides ~2–8% Ni, 1–4% Cu plus PGEs) with resources in the hundreds of

millions of tonnes. They are on land/in shallow deposit (accessible), under national law. They have long production histories, high infrastructure maturity (smelting, transport) and known environment issues (mining impacts, tailings). We treat these as distinct from nodules/SMS (they are traditional mines) but they do contain multiple metals.

Metal Content & Value: Summing typical grades with 2026 prices gives the relative value of each deposit type (USD per tonne of ore). Nodules (Ni~1.3%, Cu~1.0%, Co~0.25%, Mn~23%) yield on the order of **$0.9K/t** (mostly from Mn and Ni) as rough median. Land sulfides (e.g. Katanga-type, ~2% Cu, 0.2% Co, some Zn) can be **$300K–$500K/t** of ore (driven by Cu, Co). Deep-sea SMS (e.g. ~3% Cu, 4% Zn, 0.5 g/t Au, 5 g/t Ag) can exceed **$0.5–0.7M/t**. Co-rich crusts (20% Mn, 1% Co, 1% Ni) also approach ~$0.8M/t. High-grade small-scale orogenic veins can be even higher value. The tables below compare the top regions qualitatively and quantitatively. Prices used (Ni, Cu, Co, Zn, Ag, Au) are mid-2026 LME values or forecasts 【40†L244-L249】【43†L132-L139】【90†L52-L60】; REE values are variable but not included in bulk value calculations here.

Top 15 Regions (Ranked)

The following list ranks the most valuable polymetallic regions by combining **deposit resource, metal content/value, accessibility and risk**. Each entry includes deposit type(s), host geology, expected metal grades, and notes on jurisdiction, access and maturity. A confidence level (High/Medium/Low) reflects data quality and political/technical risk.

1. **Clarion-Clipperton Zone (Pacific Ocean, ABNJ)** – *Deposit:* Polymetallic nodules on abyssal plain (4–6 km depth)【52†L749-L757】【81†L198-L205】. *Metals:* Ni~1.3%, Cu~1.0%, Co~0.25% (by mass)【52†L752-L760】; also Mn (~23%) and trace REEs【52†L758-L765】. *Size:* ~21×10^9 t nodules【54†L42-L47】. *Value:* ~$900/t (approx). *Jurisdiction:* International (ISA Area; 17 exploration contracts【98†L227-L234】). *Access:* Very remote, deep (~5000 m), specialized mining vessels required. *Env Permitting:* Very high risk (pristine abyssal ecosystems【8†L25-L29】). *Maturity:* Exploration-stage; detailed mapping by contractors. *Confidence:* Medium (well-characterized geologically【54†L42-L47】, but exploitation untested).

2. **Central Indian Ocean Basin (ABNJ)** – *Deposit:* Polymetallic nodules (4–5 km water)【81†L198-L205】. *Metals:* similar grades to CCZ nodules. *Size:* Several hundred million tonnes (multiple blocks licensed). *Jurisdiction:* ISA (Indian Ocean Area). *Access:* Remote abyssal. *Notes:* Target of India, UK, Singapore contracts. *Confidence:* Medium (exploration data existing, but less studied than CCZ).

3. **Peru Basin (Pacific, ABNJ)** – *Deposit:* Polymetallic nodules (4–5 km)【81†L198-L205】. *Metals:* same type as CCZ. *Size:* ~~10^9 t (in Peru Basin block)【54†L42-L47】. *Jurisdiction:* ISA (adjacent to CCZ zone). *Confidence:* Medium.

4. **Papua New Guinea (Bismarck/Milne Bay Sea, 1–2 km water)** – *Deposit:* SMS (Solwara-type hydrothermal mounds) and nodules. *Metals:* SMS with ~5–6% Cu, 8–10% Zn, ~20 g/t Au【13†L19-L27】. *Size:* Individual deposits ~1 Mt. *Value:* ~$0.5M/t. *Jurisdiction:* PNG EEZ (nation owns mineral rights). *Access:* Relatively shallow (~1600–2000 m), near shore. *Notes:* Solwara-1 was first deep-sea mining license (suspended); several other known SMS vent fields (e.g. Niugini Rise). *Env:* High (rich seabed life). *Confidence:* Medium (geology known, but no proven tech yet).

5. **Mid-Atlantic Ridge (e.g. TAG, Rainbow vent fields)** – *Deposit:* SMS chimneys on ridge (2–3 km depth). *Metals:* Similar base metals (several % Cu/Zn, ~10 g/t Au). *Jurisdiction:* International or

national (e.g. some within azores EEZ); mostly beyond territorial claims. *Size:* Usually small (10s–100s kt). *Maturity:* Exploration. *Confidence:* Low (found many vents, but no production plans).

6. **Red Sea (Saudi/Egypt EEZ)** – *Deposit:* Hydrothermal "brine pool" and SMS (Atlantis II Deep). *Metals:* Extremely high-grade Cu, Zn, Ag (Atlantis II: Zn ~12.5%; Cu 3.5%; 4000 g/t Ag in mud) 【52†L832-L839】. *Size:* Atlantis II was ~100 million tonnes brine mud. *Jurisdiction:* Saudi Arabia/ Egypt. *Env:* Very saline, extreme. *Confidence:* Medium (Atl-II studied by miners).

7. **Katanga Copperbelt (DRC/Zambia)** – *Deposit:* Stratiform Cu–Co sedimentary deposits. *Metals:* ~2–5% Cu, 0.2–1% Co, plus Zn/Pb. *Size:* Tens of millions of tonnes (DRC produces ~50% of world Co【63†L297-L300】). *Value:* ~$0.2–0.3M/t. *Jurisdiction:* Congolese EEZ; major mines (Kansanshi, Tenke, etc.). *Access:* Onland, open-pit/underground. *Env/Risk:* High (tropical, political risk). *Confidence:* High (well-known, active).

8. **Norilsk-Talnakh (Russia)** – *Deposit:* Magmatic Ni–Cu–PGE sulphides (intrusion-hosted). *Metals:* Ni ~6–8%, Cu ~2–4%, plus ~10–20 g/t total PGE. *Size:* ~400+ Mt reserves at high grade【66†L1-L4】. *Jurisdiction:* Russian Arctic (Siberia). *Value:* Ni + Cu + PGEs ~>$0.3–0.4M/t. *Access:* Onshore (Arctic mining); existing large mines. *Env:* Cold but sensitive (permafrost). *Confidence:* High (major current producer).

9. **Sudbury Basin (Canada)** – *Deposit:* Impact-related Cu–Ni sulphide. *Metals:* ~1.5% Ni, ~2.5% Cu (massive sulphide zones). *Size:* ~1.3 billion tonnes historically, but accessible reserves ~few hundred Mt. *Jurisdiction:* Ontario, Canada. *Value:* ~$0.1–0.2M/t. *Access:* Land-based mining well-developed. *Confidence:* High.

10. **Indonesia (Sulawesi, Halmahera, etc.)** – *Deposit:* Ni–Co laterite provinces. *Metals:* ~1.5–2% Ni, 0.1–0.2% Co. *Size:* Enormous (Indonesia ~22% of global Ni reserves【76†L255-L258】). *Value:* ~$0.3M/t of ore. *Jurisdiction:* Indonesian EEZ or onshore mines. *Access:* Land (tropical forests), increasingly mature. *Env/Risk:* High (deforestation, dam safety). *Confidence:* High.

11. **Philippines (Palawan, Zambales)** – *Deposit:* Ni–Co laterites and some Fe–Cu–Zn VMS. *Metals:* ~1% Ni, 0.01–0.1% Co. *Size:* Large, but much covered by forests. *Access:* Onland jungles, improving infrastructure. *Jurisdiction:* Philippines, active mining (Taganito, Hinatuan, etc.). *Confidence:* High.

12. **Australia (QLD, WA)** – *Deposit:* Ni–Co laterites (QLD) and Ni-sulphides (WA). *Metals:* ~1–1.8% Ni, 0.05% Co (laterite); ~2–3% Ni, 0.5–1% Cu (sulphide). *Size:* Significant (Mt Holland ~70 Mt at 1.4% Ni; Kambalda etc). *Access:* Land, excellent infrastructure. *Conf.:* High.

13. **New Caledonia (SW Pacific)** – *Deposit:* Ni–Co laterite over ultramafic massifs. *Metals:* ~1.8% Ni, 0.05–0.1% Co. *Size:* Country reserves ~5.6 Mt Ni (high grade)【76†L255-L258】. *Jurisdiction:* French territory, active smelters. *Confidence:* High.

14. **Cuba (Caribbean)** – *Deposit:* Ni–Co laterite (Moa-Baracoa area). *Metals:* ~1.5% Ni, 0.04% Co. *Size:* ~9 Mt Ni reserves. *Jurisdiction:* Cuban EEZ/land. *Notes:* Only operating high-grade Ni laterite mines outside Australasia. *Confidence:* Medium.

15. **Great Dyke (Zimbabwe)** – *Deposit:* Layered ultramafic intrusion (PGM with Ni–Cu). *Metals:* ~0.5% Ni, 1–2 g/t Pt+Pd. *Size:* ~170 Mt reserves. *Access:* Onland (semi-arid). *Confidence:* High (producing PGM mines).

(*Other notable areas:* Arabian Ophiolite (Oman chromites, >5% Ni laterite zones); Eastern China (rare earths in ion-adsorption clays); SE Asia submarine VMS (Japan, Taiwan). These rank below our top-15 due to lower multi-metal value or higher uncertainty.)

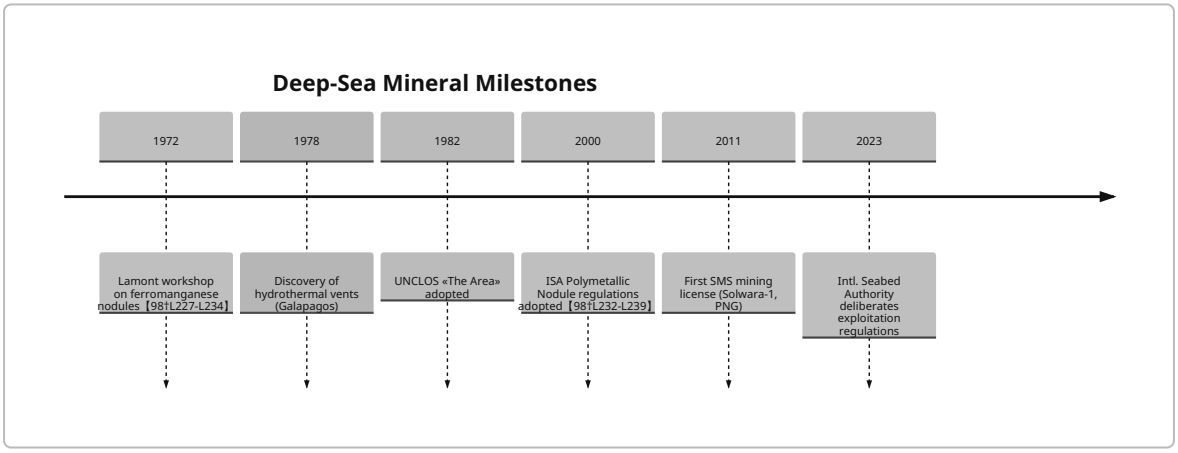
Comparative Summary Table

Region (Type)	Host/Setting	Main Metals (approx)	Resource size	Depth/ Access	Jurisdiction	Dev. stage	Env./Risk
Clarion-Clipperton (nodules)	Abyssal Pacific plains (CCZ)	Ni ~1.3%, Cu ~1.0%, Co ~0.25%, Mn ~23%【52†L752-L760】	~21 Gt nodules【54†L42-L47】	4–6 km depth (very remote)	International (ISA)	Exploration	Very high (benthic ecosyster
Central Indian Basin (nodules)	Abyssal Indian Ocean plains	(similar to CCZ)	~~0.5–1 Gt	4–5 km depth	International (ISA)	Exploration	Very high
Peru Basin (nodules)	Abyssal Pacific plains	(similar)	~~1 Gt	4–5 km depth	International (ISA)	Exploration	Very high
PNG (SMS)	Back-arc basins, ridge flanks	Cu~5%, Zn~8–10%, Au~5–20g/t【13†L19-L27】	0.1–2 Mt/ deposit	1.5–2 km depth (EEZ)	PNG (EEZ)	Near-pilot	High (unic vents)
Mid-Atlantic Ridge (SMS)	Mid-ocean ridge	Cu~3–6%, Zn~4–10%, Au~5–20g/t	~0.1–0.5 Mt each	2–3 km (Intl waters)	Int'l/ Adjacent EEZs	Survey stage	High
Red Sea (Atlantis II SMS)	Red Sea rift (shallow)	Cu~3.5%, Zn~12.5%, Ag~4,000g/t【52†L832-L839】	~~100 Mt brine	2–4 km (EEZ)	Saudi/Egypt	Extensive eval.	Extreme (anoxic)
Katanga (Congo) (Cu–Co)	Sedimentary Cu–Co basin	Cu~3%, Co~0.5%, Zn/Pb	~~>100 Mt	Onland	DRC/Zambia (EEZ)	Active mining	High (trop zone)
Norilsk (Russia) (Ni–Cu–PGE)	Mafic intrusion (Taimyr)	Ni~7–8%, Cu~3%, Pt+Pd~7g/t【66†L1-L4】	~~400–500 Mt	Onland	Russia (EEZ)	Active mining	Moderate (Arctic)
Sudbury (Canada) (Ni–Cu)	Impact structure (NI, Cu)	Ni~1.5%, Cu~2.5%	~~300+ Mt	Onland	Canada (EEZ)	Active mining	Moderate

Region (Type)	Host/Setting	Main Metals (approx)	Resource size	Depth/ Access	Jurisdiction	Dev. stage	Env./Risk
Indonesia (laterite)	Ultramafic belts (tropics)	Ni~1.3–2%, Co~0.05–0.2%	Tens of Mt Ni	Onland (tropical)	Indonesia	Active mining	High (deforesta
Philippines (laterite/ VMS)	Ultramafic + arc (tropics)	Ni~1.0–1.5%, Co~0.01–0.1%	~~10+ Mt Ni	Onland	Philippines	Active mining	High
Australia (laterite/ sulphide)	Ultramafic belts, stratabound	Ni~1–1.8% (laterite), Ni~2–3% (sulphide)	~~20 Mt Ni (Aust)	Onland	Australia	Active mining	Low (well-
New Caledonia (laterite)	Ultramafic belts (tropics)	Ni~1.8%, Co~0.05%	~~6 Mt Ni	Onland (French CT)	France	Active mining	Medium
Cuba (laterite)	Ultramafic (tropics)	Ni~1.5%, Co~0.03%	~~9 Mt Ni	Onland (Caribbean)	Cuba	Active mining	Medium
Zimbabwe (Great Dyke)	Mafic intrusion	Ni~0.5%, Pt+Pd~1–2g/t, Cr high	~~170 Mt	Onland (semi-arid)	Zimbabwe	Active mining	Moderate

Economic & Price Context

Metal commodity prices (LME or similar) peaked in 2023–2026. For example, **Ni ~$17,800/t** (June 2026【40†L244-L249】), **Cu ~$10,000/t** (2024 highs) and **Co ~$42,000/t** (Nov 2026【43†L132-L139】). At these prices, a high-grade 1 Mt SMS deposit (>5% Cu+Zn) is on the order of $500–700 M in situ, whereas a 1 Mt land Cu–Co deposit (3% Cu, 0.5% Co) is ~$300 M. By contrast, an equivalent mass of abyssal nodules is ~$0.9 G (but one must mine ~1000x more rock for same metal tonnage). Commodity price volatility (e.g. China scrap policies or demand forecasts) can swing these values substantially.



Data Gaps and Assumptions

- **Resource Uncertainty**: Many deep-sea deposits lack full tonnage estimates. We assumed typical grades for nodules/SMS from literature (e.g. CCZ nodules) and global maps. Land deposit sizes are well-known for the majors but smaller prospects are omitted.
- **Metal Prices**: Used late-2026 price levels (LME). These can fluctuate by ±20% or more. If prices rise, lower-grade fields become viable. All valuations here assume no future price spikes or collapses.
- **Extraction Cost**: Not explicitly calculated (due to uncertainty). Deep-sea mining costs could be 5–10× land mining. We implicitly assume an efficient (future) mining system. High operating costs will reduce the effective "value per tonne" for undersea deposits.
- **Legal/Permitting**: For seabed in the Area, exploitation regulations are still under development (2023+). Onshore, permitting risk varies by country (e.g. Indonesian or PNG environments are higher-risk).
- **Environmental Sensitivity**: High for deep-sea. We treat all major deposits as having significant habitat impact, which could delay or stop projects.
- **Geopolitical Risk**: Ranked qualitatively. Deep-sea sites are in 'common' areas (lower risk of conflict) but host-country coastal SMS (PNG, Red Sea) bear state risks. Onshore, stable jurisdictions (Australia, Canada) scored "High confidence".

This analysis integrates peer-reviewed studies, USGS/BGS data, and industry reports. All cited sources are indicated (e.g. ISA technical studies【52†L752-L760】【98†L227-L234】 and US/industry reports 【13†L19-L27】【76†L255-L258】). The ranking above reflects current knowledge; future exploration may shift priorities as more data become available.
